

Lesson 5-1 · Single-Period Quick

n th Roots, Rational Exponents + Cylinder
DOK-3 (Item 1A Prep)

Klimsara-adapted · Student-centered · No Blooket

Do Now — $y = x^2$ graph: roots from squares

Algebra 2 · Lesson 5-1

DOK 1

5 min

bridge to n th roots

Do Now — Explore & Reason: $y = x^2$ Graph of $y = x^2$ is projected. For each, find the missing value:

- (a) (2, --) (b) (3, --) (c) (-3, --) (d) (5, --)
(e) (--, 4) (f) (--, -16) (g) (--, 7)

B. How do you find $\sqrt{13}$ using this graph?

C. Sketch $y = x^3$. Approximate $\sqrt[3]{5}$, $\sqrt[3]{-5}$, $\sqrt[3]{8}$.

Launch — Example 1 + Example 3 (two examples, compressed)

Algebra 2 · Lesson 5-1

DOK 1-2

13 min

teacher models

n th Roots $\leftrightarrow$ Rational Exponents — Rules Card

1. $x^{1/n} = \sqrt[n]{x}$ (principal n th root)
2. $x^{m/n} = (\sqrt[n]{x})^m = \sqrt[n]{x^m}$
3. Rationalize n th-root denominator: multiply by $\sqrt[n]{x^{n-1}}$
4. Even index: radicand ≥ 0 in Reals; odd index: any real

Example 1: real n th roots — **Example 3:** evaluate rational exponents

Release to Explore at minute 18. Copy Rules card into margin.

Explore — TPS items 1–4 (first 14 min) Algebra 2 .

Lesson 5-1

DOK 2–3

32 min

student work

7 items in order. Plan ~10 min for Practice #50.

Try It 1 · Real nth roots Find the real fourth roots of 81; the real cube roots of 64.

Try It 3 · Evaluate rational exponents Evaluate $-(16^{3/4})$ and $\sqrt[5]{3.5^4}$. Round if needed.

Practice #28 · nth roots with variables Find the real square roots of 25.

Practice #30 · Rational exponent form Rewrite $\sqrt[6]{729}$ using a fractional exponent.

Explore — TPS Items 3–7 + DOK-3 Capstone (18 min)

Algebra 2 · Lesson 5-1

DOK 2–3

32 min

student work

Practice #37 · Simplify: mixed radical/rational Simplify

$\sqrt[6]{729a^{24}b^{18}}$. Assume all variables positive.

Practice #48 · Multi-step rational exponent Which ex-

pressions equal $b^{3/4}$? (Yes/No table — see packet.)

Practice #50 · CYLINDER MODELING ★ Cylindrical

container: height = diameter. Volume = 169.65 ft^3 .

Part A: lateral surface area? **Part B:** containers in 20-ft hold?

Set up $V = \pi r^2(2r)$, isolate r in rational-exponent form, then rationalize.

Share / Summary — Cylinder Reveal

Algebra 2 .

Lesson 5-1

DOK 2

5 min

DOK-3 reveal + assessment bridge

Sentence frame · Cylinder Capstone “I isolated r from $V = \pi r^2 h$ by substituting $h = 2r$, giving $V = \underline{\hspace{2cm}}$. Solving for r , I got $r = \underline{\hspace{2cm}}$. In rational-exponent form, $r = (V/(2\pi))^{1/3}$. To rationalize I multiplied by $\underline{\hspace{1cm}}$ to get $\underline{\hspace{2cm}}$.”

Bridge to Topic 5 Assessment — Item 1A Prep The cylinder move: isolate a variable $\rightarrow$ rational-exponent form $\rightarrow$ rationalize. This is **exactly** what Topic 5 Assessment Item 1A asks. You have now rehearsed it.

Exit Ticket — Summary Recap

Algebra 2 · Lesson 5-1

DOK 1–2

3 min

not graded

Complete on your exit ticket

1. $x^{m/n}$ is the same as _____ in radical form.
2. To rationalize a cube-root denominator I _____.
3. One biggest thing I learned today: _____.